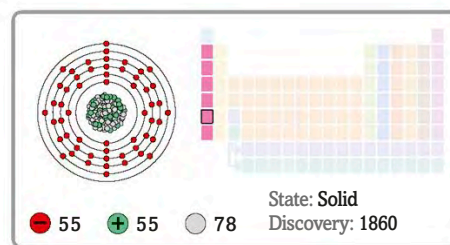


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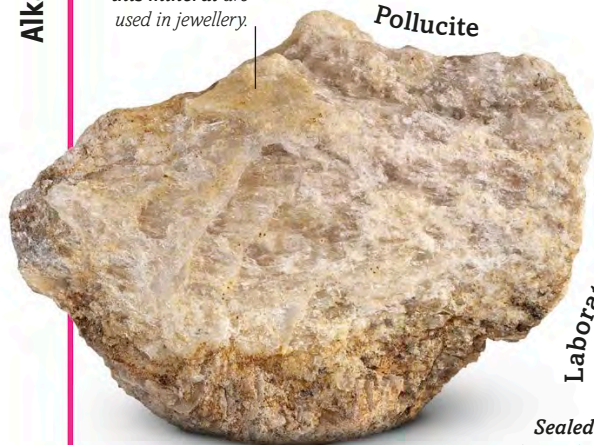
Caesium



Forms

The crystals of this mineral are used in jewellery.

Pollucite



Shiny, silver-gold metal

Laboratory sample of pure caesium in an airless vial

Sealed glass tube



KIRCHHOFF AND BUNSEN

Caesium was discovered in 1860 by German scientists Robert Bunsen and Gustav Kirchhoff. They burned a sample of mineral water on a burner, which split the flame's light into individual colours. One of them was a distinctive light blue, which came from caesium.



Gustav Kirchhoff (left) and Robert Bunsen (right)

Uses

This highly accurate clock is also called a caesium clock.



Atomic clock



High-density caesium compounds in this fluid stop toxic gases rising to the surface.

Drilling fluid

As the most reactive metal on Earth, caesium explodes into flames if in contact with air or water. Therefore, **pure caesium**, is stored in a sealed glass tube from which all the air has been sucked out. This element is rare, and most of it is extracted from the mineral **pollucite**. Its name

means "sky blue" and refers to the colour of caesium's flame when burning. Caesium is used in **atomic clocks**, which measure time down to a billionth of a second. These clocks are so accurate that they would gain or lose no more than one second every 300 years.